

DSA 150 Pattern Guide

DSA Pattern Identification Guide

A reference for spotting the right pattern (and subtype) when you see a new problem.

Step 1 — Input Structure → First Guess

Input Looks Like	First Think
Array + find/count something	HashMap / Two Pointers / Sliding Window
Sorted array	Binary Search / Two Pointers
String / substring	Sliding Window / HashMap
Linked list	Fast-Slow / Reversal / Pointer manipulation
Tree	DFS / BFS
Grid	DFS / BFS
Graph	DFS / BFS / Topological Sort / Union-Find
“Top K”, “Kth largest”	Heap
“All possible...”	Backtracking
Prefix / dictionary words	Trie
“Maximum/minimum ways”	DP / Greedy
Intervals [start, end]	Sorting + Intervals / Greedy
Bits / XOR	Bit Manipulation

Step 2 — Master Pattern Table (with Subtypes)

#	Pattern	Subtype(s)	Trigger Words / Clues	Input Signal	Example
1	HashMap / HashSet	Frequency-based, Lookup-based, Set-based, Index Mapping, Grouping	“duplicate”, “frequency”, “count occurrences”, “seen before”, “unique”	Array, need to find/count something	Two Sum, Contains Duplicate, Valid Anagram, Group Anagrams, Top K Frequent
2	Two Pointers	Opposite ends (L+R), Same direction (fast/slow), Partition / Dutch flag	sorted array, palindrome, pair/triplet, left/right, opposite ends	Sorted array or compare two positions	Valid Palindrome, Two Sum II, 3Sum, Container With Most Water, Trapping Rain Water
3	Sliding Window	Fixed size, Variable size, Expand-Shrink, Monotonic Window (deque)	“longest/shortest substring”, “max sum subarray”, “at most K”, “contiguous”	Best contiguous part of array/string	Longest Substring w/o Repeating (variable), Permutation in String (fixed), Sliding Window Max (monotonic deque)
4	Stack / Monotonic Stack	Increasing stack, Decreasing stack, Nearest Greater/Smaller, Min/Max Stack, Expression Handling, Histogram/Span	parentheses, undo, “next greater/smaller”, nested structure	Need to track most recent unresolved item	Daily Temperatures (decreasing), Largest Rectangle (increasing)
5	Binary Search	On Index (classic), On Answer (feasibility check), On Partition	“minimum/maximum possible X”, “smallest speed/capacity”	Monotonic search space, can eliminate half	Koko Eating Bananas (on answer), rotated array search (on index), Median of Two Sorted Arrays (on partition)
6	Fast & Slow Pointer	Cycle Detection, Finding Middle	cycle, middle of list, relative position	Linked list needing cycle/midpoint detection	Linked List Cycle, Find the Duplicate Number
7	Linked List Reversal	Full Reverse, Partial (k-group)	“reverse”, “reverse every K nodes”, “reorder”	Pointer manipulation task	Reverse Linked List (full), Reverse Nodes in k-Group (partial)
8	Tree DFS	Preorder, Inorder, Postorder, Top-Down recursion, Bottom-Up recursion	depth, height, diameter, path, subtree	Tree, need to go deep first	Max Depth (bottom-up), Invert Tree (top-down), Diameter (path-based)
9	Tree BFS	Level Order, Zigzag, Right Side View	level, level-by-level, shortest distance (unweighted)	Tree/graph, need level-wise processing	Level Order Traversal, Right Side View
10	Heap / Priority Queue	Top-K, Kth Element, K-Closest, Two-Heap	“K largest/smallest”, “Top K”, median	Need efficient access to extremum	Kth Largest (top-K), Median from Data

		(median), K-way Merge, Greedy+Heap			Stream (two-heap), Merge k Sorted Lists (K-way merge), Task Scheduler (greedy+heap)
11	Backtracking	Subsets (power set), Permutations/Combinations, Word Search on Grid, Palindrome Partitioning, Pruning/State Tracking	“all possible”, “generate all”, “every combination/permutation”	Decision tree: choose-explore-unchoose	Subsets, Permutations, Word Search, N-Queens (pruning)
12	Trie	Insert/Search, Prefix Match, Bitwise Trie	prefix, dictionary, autocomplete	Repeated prefix matching across words	Implement Trie, Word Search II (trie+backtrack)
13	Grid DFS/BFS	Flood Fill, Multi-source BFS, Boundary-first Marking	islands, connected cells, regions, spread	2D grid, connected-cell exploration	Number of Islands (flood fill), Rotting Oranges (multi-source BFS), Surrounded Regions (boundary-first)
14	Topological Sort	Kahn’s Algorithm (BFS/in-degree), DFS-based Topo Sort	prerequisites, dependencies, “course schedule”	Directed graph with ordering constraints	Course Schedule II (Kahn’s), Alien Dictionary (DFS-based)
15	Union-Find / DSU	Basic Union-Find, Union by Rank/Size, Cycle Detection (undirected)	connected components, redundant connection, groups	Dynamic connectivity tracking	Redundant Connection, Graph Valid Tree
16	Shortest Path	BFS (unweighted), Dijkstra (weighted, non-negative), Bellman-Ford (weighted, negative edges/constraints), Floyd-Warshall (all-pairs)	minimum distance/cost/time A→B	Depends on weighted vs unweighted	Word Ladder (BFS), Network Delay Time (Dijkstra), Cheapest Flights K Stops (Bellman-Ford)
17	Dynamic Programming	Memoization (top-down), Tabulation (bottom-up), Linear DP, Grid DP, Decision DP	Repeats + breaks into smaller subproblems	Overlapping subproblems + optimal substructure	Climbing Stairs (linear), Unique Paths (grid)
18	Knapsack (DP)	0/1 Knapsack, Unbounded Knapsack, Subset Sum	choose items, target sum, “can we make X?”, use/not use	Take-or-skip decision per element	Partition Equal Subset Sum (0/1), Coin Change (unbounded)
19	String DP	Sequence DP (LCS-style), Edit/Transform DP, Interval DP (palindromes)	two strings, transform, subsequence, edit, interleave	2D DP over string indices	LCS, Edit Distance (sequence), Palindromic Substrings (interval)
20	Greedy	Interval Greedy, Scheduling (deadline/profit-based), Resource Allocation, Jump Game pattern	choose earliest/largest/smallest, “reach farthest”	Locally optimal choice → global optimum	Jump Game (reach-farthest), Gas Station, Non-overlapping Intervals
21	Intervals	Merge, Insert, Overlap Detection, Sweep Line (min rooms/platforms)	Input is [start, end] pairs	Sort by start, check overlap	Merge Intervals, Meeting Rooms II (sweep line)
22	Matrix Manipulation	In-place Transform (transpose+reverse), Boundary/Spiral Traversal, Marking-based (zeroing)	rotate, spiral, set rows/cols to zero	2D matrix, in-place transformation	Rotate Image (transpose+reverse), Spiral Matrix (boundary), Set Matrix Zeroes (marking)
23	Bit Manipulation	XOR Pattern, Bit Masking, Bit Checks (Kernighan’s), Subset-via-Bits	XOR, bits, set bits, “number of 1s”	Numeric problem with binary structure	Single Number (XOR), Number of 1 Bits (Kernighan’s), Sum of Two Integers (bit masking)

Golden Trigger Phrases (fast lookup)

Phrase in Problem	Pattern
“Top K”	Heap — first instinct
“Return all possible...”	Backtracking
“Prerequisite”	Topological Sort
“Can I guess X and check feasibility?”	Binary Search on Answer
Repeats + breaks into smaller same-shape problems	DP
“take it or skip it”	Knapsack DP
Two strings, transform/match	String DP

How to Use This

1. Read the problem. Identify the **input structure** (Step 1 table) for a first guess.
2. Scan for **trigger words** in the problem statement (Step 2 table / Golden Phrases).
3. Narrow down to the **subtype** — this determines the exact template/technique to apply (e.g. “Sliding Window” → is it fixed size or variable size?).
4. Cross-check against the [Ultimate DSA 150 tracker](#) for a worked example of that exact pattern.

Companion file: [Ultimate_DSA_150_Patterns_and_Links.md](#) — all 150 questions tagged with pattern, LeetCode link, and YouTube search link.

Ultimate DSA 150 — Pattern + Link Tracker

Source: *Ultimate DSA 150 by Mayank* (@themonkeystalk). LC links go to the actual LeetCode problem page. YT links are YouTube **search-query** links (not fixed videos) — that’s how they were embedded in the original PDF.

Arrays & Hashing

#	Question	Pattern	LeetCode	YouTube
1	Two Sum	Hash Map (lookup-based)	Link	Search
2	Contains Duplicate	Hash Set (existence check)	Link	Search
3	Valid Anagram	Hash Map (frequency count)	Link	Search
4	Group Anagrams	Hash Map (grouping)	Link	Search
5	Top K Frequent Elements	Hash Map + Bucket Sort	Link	Search
6	Product of Array Except Self	Prefix/Suffix Product	Link	Search
7	Valid Sudoku	Hash Set (index mapping)	Link	Search
8	Encode and Decode Strings	Length-prefix Encoding (design)	Link	Search
9	Longest Consecutive Sequence	Hash Set (sequence start)	Link	Search

Two Pointers

#	Question	Pattern	LeetCode	YouTube
10	Valid Palindrome	Two Pointer (opposite ends)	Link	Search
11	Two Sum II	Two Pointer (sorted array)	Link	Search
12	3Sum	Two Pointer + Sort	Link	Search
13	Container With Most Water	Two Pointer (greedy shrink)	Link	Search
14	Trapping Rain Water	Two Pointer (prefix max)	Link	Search

Sliding Window

#	Question	Pattern	LeetCode	YouTube
15	Best Time to Buy and Sell Stock	Sliding Window (single pass)	Link	Search
16	Longest Substring Without Repeating Characters	Sliding Window (variable)	Link	Search
17	Longest Repeating Character Replacement	Sliding Window (variable)	Link	Search
18	Permutation in String	Sliding Window (fixed size)	Link	Search
19	Minimum Window Substring	Sliding Window (expand-shrink)	Link	Search
20	Sliding Window Maximum	Monotonic Deque	Link	Search

Stack / Monotonic Stack

#	Question	Pattern	LeetCode	YouTube
21	Valid Parentheses	Stack (matching pairs)	Link	Search
22	Min Stack	Stack (auxiliary min)	Link	Search
23	Evaluate Reverse Polish Notation	Stack (expression eval)	Link	Search
24	Generate Parentheses	Backtracking	Link	Search
25	Daily Temperatures	Monotonic Stack (decreasing)	Link	Search
26	Largest Rectangle in Histogram	Monotonic Stack (increasing)	Link	Search

Binary Search

#	Question	Pattern	LeetCode	YouTube
27	Binary Search	Binary Search (classic)	Link	Search
28	Search a 2D Matrix	Binary Search (flattened)	Link	Search
29	Koko Eating Bananas	Binary Search on Answer	Link	Search
30	Search in Rotated Sorted	Binary Search (modified)	Link	Search

	Array			
31	Find Minimum in Rotated Sorted Array	Binary Search (pivot find)	Link	Search
32	Time Based Key-Value Store	Binary Search (design)	Link	Search
33	Median of Two Sorted Arrays	Binary Search on Partition	Link	Search

Linked List

#	Question	Pattern	LeetCode	YouTube
34	Reverse Linked List	Pointer Reversal (full)	Link	Search
35	Merge Two Sorted Lists	Two Pointer Merge	Link	Search
36	Reorder List	Fast-Slow + Reverse + Merge	Link	Search
37	Remove Nth Node From End of List	Two Pointer (gap)	Link	Search
38	Copy List with Random Pointer	Hash Map (node mapping)	Link	Search
39	Add Two Numbers	Simulation (carry)	Link	Search
40	Linked List Cycle	Fast-Slow (Floyd's)	Link	Search
41	Find the Duplicate Number	Fast-Slow (Floyd's)	Link	Search
42	LRU Cache	Hash Map + Doubly Linked List	Link	Search
43	Merge k Sorted Lists	Heap (K-way merge)	Link	Search
44	Reverse Nodes in k-Group	Pointer Reversal (partial)	Link	Search

Trees / DFS on Trees

#	Question	Pattern	LeetCode	YouTube
45	Invert Binary Tree	DFS (top-down)	Link	Search
46	Maximum Depth of Binary Tree	DFS (bottom-up)	Link	Search
47	Diameter of Binary Tree	DFS (path-based)	Link	Search
48	Balanced Binary Tree	DFS (bottom-up height)	Link	Search
49	Same Tree	DFS (simultaneous)	Link	Search
50	Subtree of Another Tree	DFS (nested comparison)	Link	Search
51	Lowest Common Ancestor of a BST	BST Recursion	Link	Search
52	Binary Tree Level Order Traversal	BFS (level order)	Link	Search
53	Binary Tree Right Side View	BFS (last per level)	Link	Search
54	Count Good Nodes in Binary Tree	DFS (top-down max-so-far)	Link	Search
55	Validate Binary Search Tree	DFS (bounds-passing)	Link	Search
56	Kth Smallest Element in a BST	DFS (inorder)	Link	Search
57	Construct Binary Tree from Preorder and Inorder Traversal	DFS + Hash Map	Link	Search
58	Binary Tree Maximum Path Sum	DFS (path-based, bottom-up)	Link	Search
59	Serialize and Deserialize Binary Tree	DFS (preorder, design)	Link	Search

Heap / Priority Queue

#	Question	Pattern	LeetCode	YouTube
60	Kth Largest Element in a Stream	Heap (min-heap size K)	Link	Search
61	Last Stone Weight	Heap (max-heap simulation)	Link	Search
62	K Closest Points to Origin	Heap (Top-K)	Link	Search
63	Kth Largest Element in an Array	Heap / Quickselect	Link	Search
64	Task Scheduler	Greedy + Heap	Link	Search
65	Design Twitter	Heap (K-way merge) + Hash Map	Link	Search
66	Find Median from Data	Two Heaps	Link	Search

Backtracking

#	Question	Pattern	LeetCode	YouTube
67	Subsets	Backtracking (power set)	Link	Search
68	Subsets II	Backtracking (power set + dedup)	Link	Search
69	Combination Sum	Backtracking (unbounded)	Link	Search
70	Combination Sum II	Backtracking (bounded + dedup)	Link	Search
71	Permutations	Backtracking (permutation)	Link	Search
72	Permutations II	Backtracking (permutation + dedup)	Link	Search
73	Letter Combinations of a Phone Number	Backtracking (Cartesian product)	Link	Search
74	Word Search	Backtracking (grid DFS)	Link	Search
75	Palindrome Partitioning	Backtracking (partition)	Link	Search
76	N-Queens	Backtracking (constraint placement)	Link	Search

Trie

#	Question	Pattern	LeetCode	YouTube
77	Implement Trie	Trie (insert/search)	Link	Search
78	Design Add and Search Words Data Structure	Trie + DFS (wildcard)	Link	Search
79	Word Search II	Trie + Backtracking	Link	Search

Graphs / BFS / DFS

#	Question	Pattern	LeetCode	YouTube
80	Number of Islands	DFS/BFS (connected components)	Link	Search
81	Clone Graph	DFS + Hash Map	Link	Search
82	Max Area of Island	DFS (area accumulation)	Link	Search
83	Pacific Atlantic Water Flow	Multi-source DFS/BFS	Link	Search
84	Surrounded Regions	DFS (boundary-first)	Link	Search
85	Rotting Oranges	Multi-source BFS	Link	Search
86	Walls and Gates	Multi-source BFS	Link	Search
87	Course Schedule	Cycle Detection (Directed)	Link	Search
88	Course Schedule II	Topological Sort	Link	Search
89	Redundant Connection	Union-Find (DSU)	Link	Search
90	Number of Connected Components in an Undirected Graph	Union-Find (DSU)	Link	Search
91	Graph Valid Tree	Union-Find / Cycle Detection	Link	Search
92	Word Ladder	BFS (shortest path, implicit graph)	Link	Search

Advanced Graphs

#	Question	Pattern	LeetCode	YouTube
93	Reconstruct Itinerary	Eulerian Path (Hierholzer's)	Link	Search
94	Network Delay Time	Dijkstra	Link	Search
95	Cheapest Flights Within K Stops	Bellman-Ford	Link	Search
96	Swim in Rising Water	Binary Search on Answer + BFS / Dijkstra-variant	Link	Search
97	Alien Dictionary	Topological Sort	Link	Search
98	Critical Connections in a Network	Bridges (Tarjan's)	Link	Search

1-D Dynamic Programming

#	Question	Pattern	LeetCode	YouTube
99	Climbing Stairs	DP (Fibonacci-style)	Link	Search
100	Min Cost Climbing Stairs	DP (linear decision)	Link	Search
101	House Robber	DP (skip/take)	Link	Search
102	House Robber II	DP (circular array split)	Link	Search
103	Coin Change	DP (unbounded knapsack, min)	Link	Search
104	Coin Change II	DP (unbounded knapsack, count)	Link	Search
105	Decode Ways	DP (sequence decision)	Link	Search
106	Longest Increasing Subsequence	DP (sequence) / Binary Search	Link	Search
107	Partition Equal Subset Sum	DP (0/1 knapsack)	Link	Search
108	Word Break	DP (sequence segmentation)	Link	Search
109	Maximum Product Subarray	DP (running max & min)	Link	Search
110	Jump Game	Greedy / DP (reachability)	Link	Search

2-D Dynamic Programming

#	Question	Pattern	LeetCode	YouTube
111	Unique Paths	DP (grid)	Link	Search
112	Unique Paths II	DP (grid, obstacles)	Link	Search
113	Minimum Path Sum	DP (grid)	Link	Search
114	Triangle	DP (grid, bottom-up)	Link	Search
115	Longest Common Subsequence	DP (2D sequence)	Link	Search
116	Edit Distance	DP (2D sequence)	Link	Search
117	Interleaving String	DP (2D sequence)	Link	Search
118	Distinct Subsequences	DP (2D sequence, counting)	Link	Search
119	Palindromic Substrings	DP (interval) / Expand-Center	Link	Search
120	Longest Palindromic Subsequence	DP (interval)	Link	Search
121	Burst Balloons	DP (interval, partition-style)	Link	Search

Greedy

#	Question	Pattern	LeetCode	YouTube
122	Maximum Subarray	Kadane’s Algorithm	Link	Search
123	Best Time to Buy and Sell Stock II	Greedy (positive diffs)	Link	Search
124	Jump Game	Greedy (max reach)	Link	Search
125	Jump Game II	Greedy (BFS-style level jump)	Link	Search
126	Gas Station	Greedy (total sum + reset)	Link	Search
127	Non-overlapping Intervals	Greedy (min removals)	Link	Search
128	Partition Labels	Greedy (last-occurrence)	Link	Search
129	Valid Parenthesis String	Greedy (range tracking)	Link	Search

Intervals

#	Question	Pattern	LeetCode	YouTube
130	Insert Interval	Interval Merge (linear scan)	Link	Search
131	Merge Intervals	Interval Merge (sort + merge)	Link	Search
132	Non-overlapping Intervals	Greedy (min removals)	Link	Search
133	Meeting Rooms	Interval Sort + Overlap Check	Link	Search
134	Meeting Rooms II	Heap (min-heap end times)	Link	Search
135	Interval List Intersections	Two Pointer (interval intersect)	Link	Search

Math / Geometry

#	Question	Pattern	LeetCode	YouTube
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136	Rotate Image	In-place Matrix Transform	Link	Search
137	Spiral Matrix	Matrix Traversal (boundary sim)	Link	Search
138	Set Matrix Zeroes	Matrix (in-place marking)	Link	Search
139	Happy Number	Fast-Slow (cycle detection)	Link	Search
140	Pow(x, n)	Divide & Conquer (fast exponent)	Link	Search
141	Plus One	Array Simulation (carry)	Link	Search
142	Count Primes	Sieve of Eratosthenes	Link	Search
143	Missing Number	Bit Manipulation (XOR) / Math	Link	Search

Bit Manipulation

#	Question	Pattern	LeetCode	YouTube
144	Single Number	XOR Pattern	Link	Search
145	Single Number II	Bit Masking (mod 3 count)	Link	Search
146	Number of 1 Bits	Bit Checks (Brian Kernighan's)	Link	Search
147	Counting Bits	DP + Bit Masking	Link	Search
148	Sum of Two Integers	Bit Masking (XOR + AND carry)	Link	Search
149	Reverse Bits	Bit Masking (bit-by-bit)	Link	Search
150	Bitwise AND of Numbers Range	Bit Masking (common prefix shift)	Link	Search

Note: #110 and #124 are the same LeetCode problem (Jump Game) listed twice in the original PDF — once under 1-D DP, once under Greedy, since it can be solved either way. Same for #127/#132 (Non-overlapping Intervals).